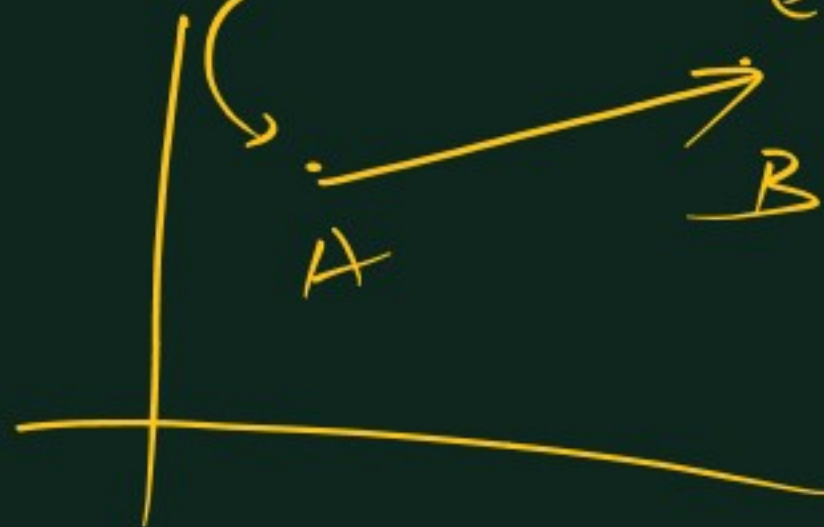


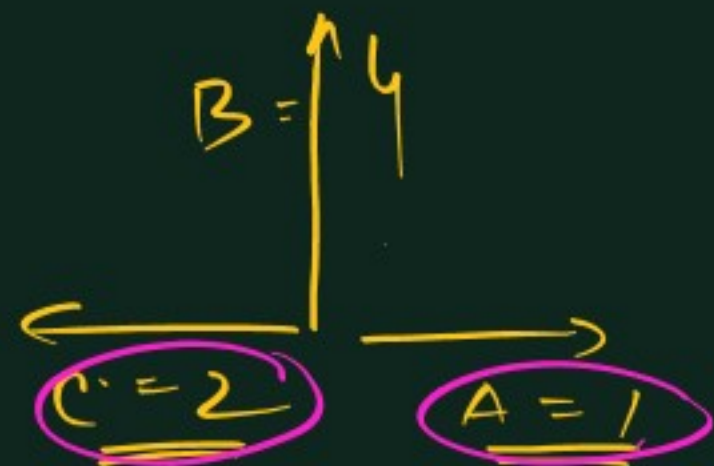
## Doubts

38)  $(4, -4, 0)$   $(-2, -2, 0)$



$$\sqrt{(4 - (-2))^2 + (-4 - (-2))^2 + (0 - 0)^2}$$
$$= \underline{\underline{2\sqrt{10}}}$$

64)



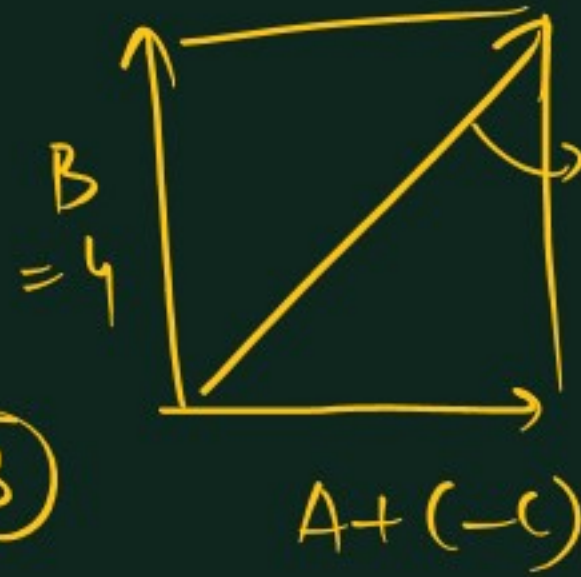
$$\boxed{A - C} - B$$

$$A \oplus (-C)$$

$$A = 1$$

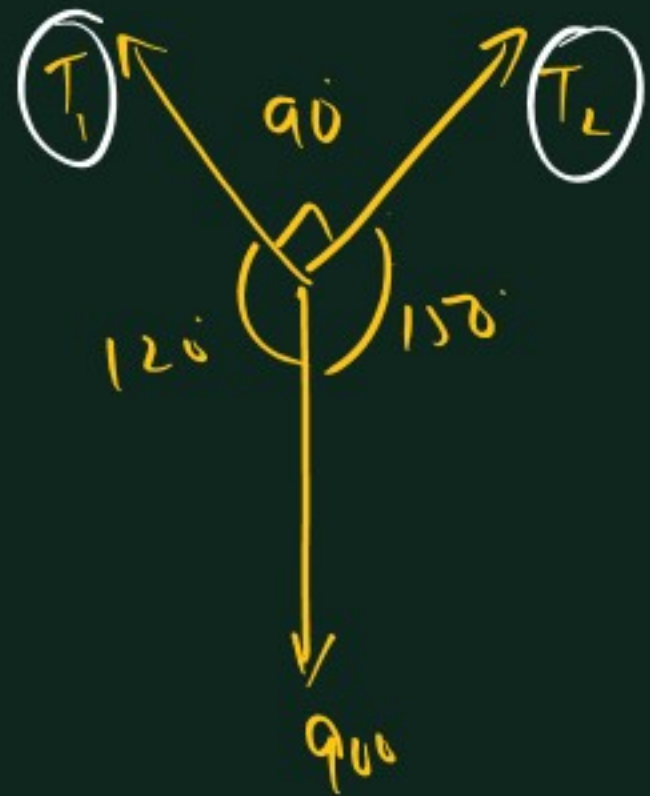
$$\frac{2}{-C}$$

} ③



→ Resolution  
→ D.P.

5) Equi<sup>m</sup>  $\Rightarrow \Sigma F = 0$



Lami's Theorem

$$\frac{T_1}{\sin 150} = \frac{T_2}{\sin 120} = \frac{900}{\sin 90}$$

$\swarrow$   $\searrow$   
 $\sin(180-30)$   $+\frac{\sqrt{3}}{2}$   
 $= +\frac{1}{2}$

$$\Rightarrow \frac{T_1}{(1/2)} = \frac{T_2}{\sqrt{3}/2} \Rightarrow \frac{T_1}{T_2} = \frac{1/2}{\sqrt{3}/2} = \frac{1}{\sqrt{3}}$$

6)  $F_1, F_2$

$$\Rightarrow \frac{F_1 + F_2}{F_1 - F_2} = \frac{7}{3}$$

$$\Rightarrow 3F_1 + 3F_2 = 7F_1 - 7F_2$$

$$\Rightarrow 10F_2 = 4F_1$$

$$\Rightarrow \frac{F_1}{F_2} = \frac{10}{4} = \frac{5}{2}$$



$$66) \begin{cases} 1 + \cos \theta = 2 \cos^2(\frac{\theta}{2}) \\ 1 - \cos \theta = 2 \sin^2(\frac{\theta}{2}) \end{cases}$$

$$\cos 2\theta = 2 \cos^2 \theta - 1$$

If 2 equal magnitude  
vectors ( $|\vec{A}| = A$ ,  
 $|\vec{B}| = A$ )

$$|\vec{A} + \vec{B}| = 2A \cos(\theta/2)$$

$$|\vec{A} - \vec{B}| = 2A \sin(\theta/2)$$

$$|\vec{P}_1 - \vec{P}_2| = \sqrt{P^2 + P^2 - 2PP \cos \theta}$$

$$= \sqrt{2P^2 (1 - \cos \theta)}$$

$\hookrightarrow 2 \sin^2 \theta/2$

$$= 2P \sin \theta/2$$

## Unit vector

→ Magnitude = 1

→ Represent direction

→  $\hat{a}, \hat{n}, \hat{i} \dots$  ← Cap symbol

→  $\left\{ \begin{array}{l} \hat{i} \Rightarrow x\text{-direction} \\ \hat{j} \Rightarrow y\text{-direction} \\ \hat{k} \Rightarrow z\text{-direction} \end{array} \right\}$



$$\vec{F}_1 = 10 \hat{i} \quad 5 \downarrow \vec{F}_2$$

$$\vec{F}_2 = 5(-\hat{j})$$

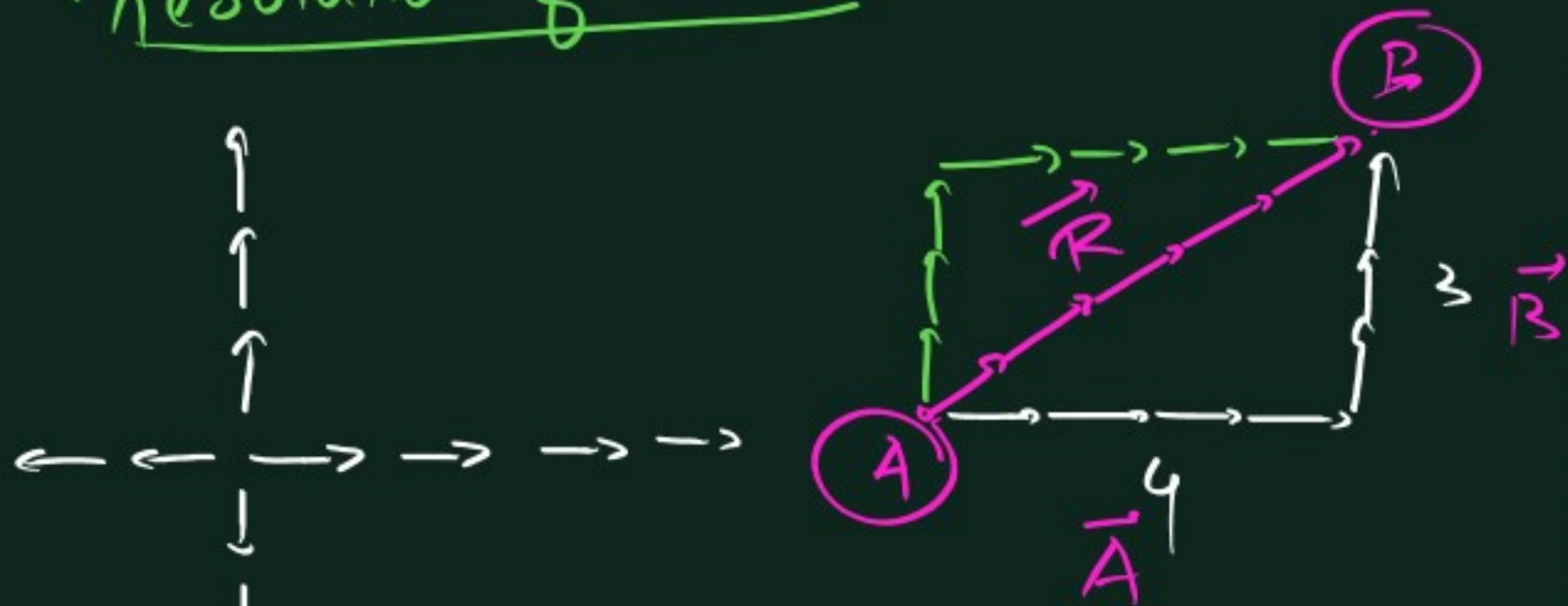
→ In general,  $\vec{A} = \text{Magnitude} \times \text{Direction}$

FORMULA  
To obtain unit vector  $\Rightarrow$

$$\vec{A} = |\vec{A}| \times \hat{A}$$
$$\hat{A} = \frac{\vec{A}}{|\vec{A}|}$$



# Resolution of vector



$$\vec{R} = \vec{A} + \vec{B}$$

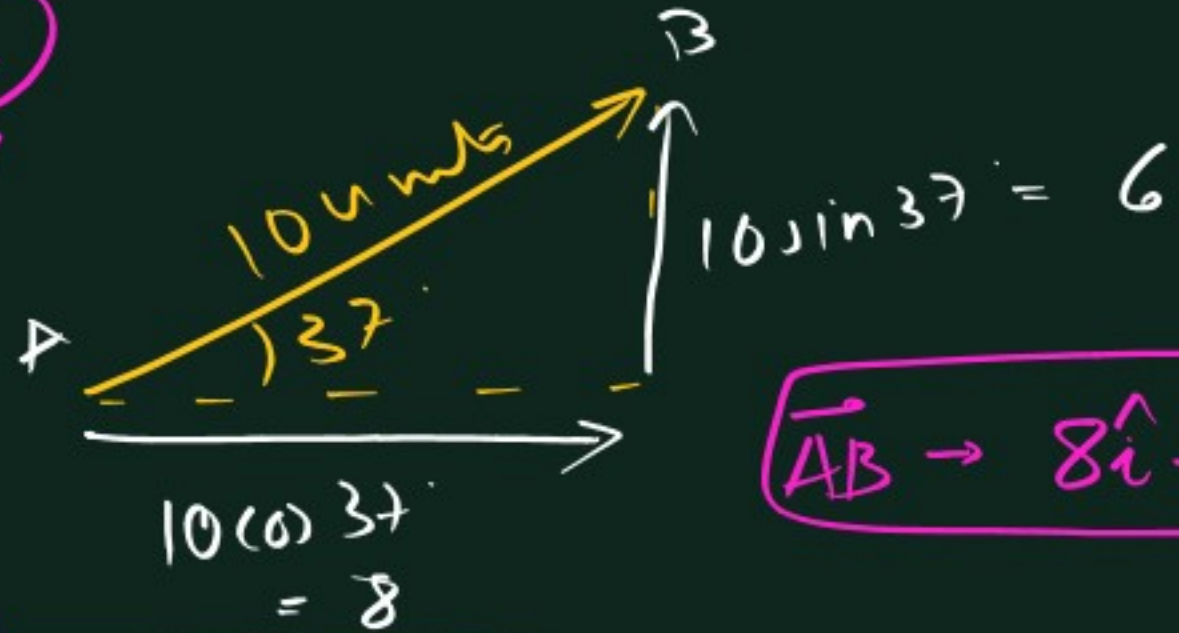
Diagram showing the addition of two vectors  $\vec{A}$  and  $\vec{B}$  to find their resultant  $\vec{R}$  using the triangle rule. Vector  $\vec{A}$  is horizontal, and vector  $\vec{B}$  is vertical. The resultant  $\vec{R}$  is the hypotenuse of the right-angled triangle formed by  $\vec{A}$  and  $\vec{B}$ . The points A, B, and R are circled in pink.



$$P = H \sin \theta$$

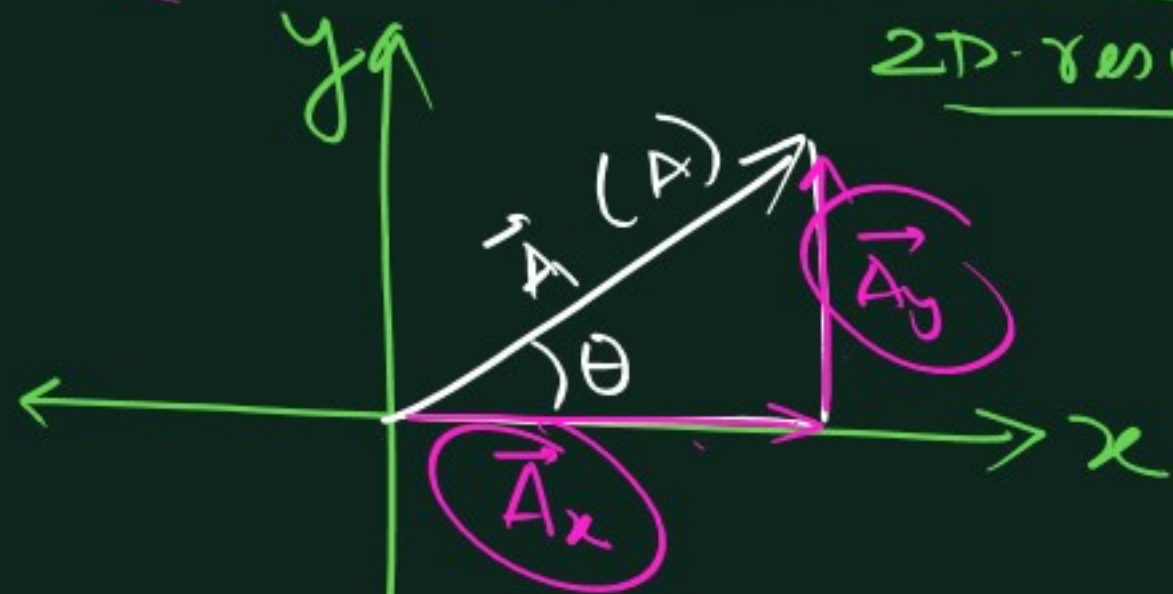
$$B = H \cos \theta$$

Ex.



$$\vec{AB} \rightarrow 8\hat{i} + 6\hat{j}$$

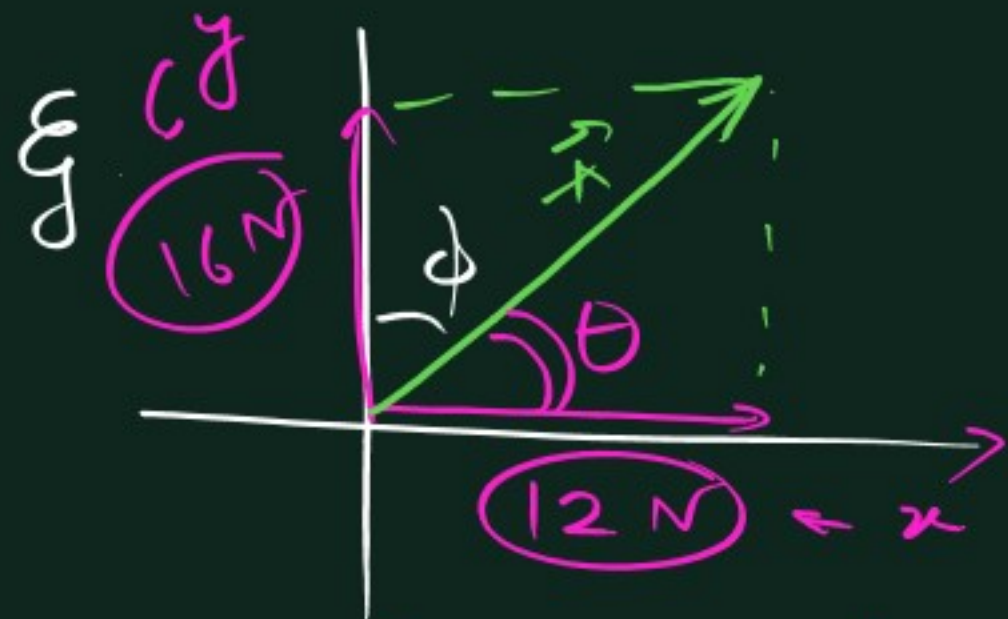
2D-resolution



$$\vec{A} = \vec{A}_x + \vec{A}_y$$

$$\vec{A} = A \cos \theta \hat{i} + A \sin \theta \hat{j}$$

Components  $\Rightarrow$  vector  $(\vec{A}), \theta$



Find  $|\vec{A}|$  &  $\theta$  with x axis.

$$|\vec{A}| = \sqrt{16^2 + 12^2} = 20$$

$$\tan \theta = \frac{16}{12} = \frac{4}{3} \Rightarrow \theta = 53^\circ$$

$\rightarrow \underline{A_x \text{ \& } A_y}$

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2}$$

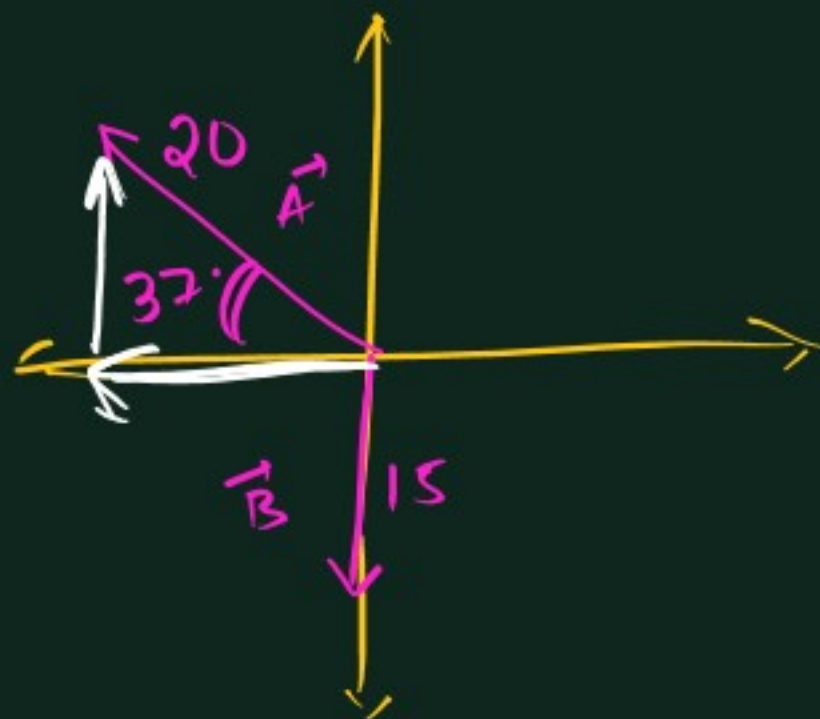
$$\tan \theta = \frac{A_y}{A_x}$$

$\theta$  = angle with x-axis.

$$\vec{A} = A_x \hat{i} + A_y \hat{j}$$



Q1.

Find (in terms of  $\hat{i}, \hat{j}$ )

i)  $\vec{A}$

ii)  $\vec{B}$

iii)  $\vec{A} + \vec{B}$  //

$$\begin{aligned}\vec{A} &= 20 \cos 37^\circ (-\hat{i}) + 20 \sin 37^\circ (\hat{j}) \\ &= 16(-\hat{i}) + 12\hat{j} \\ &= -16\hat{i} + 12\hat{j}\end{aligned}$$

$$\begin{aligned}\vec{B} &= 0\hat{i} + 15(-\hat{j}) \\ &= 0\hat{i} - 15\hat{j}\end{aligned}$$



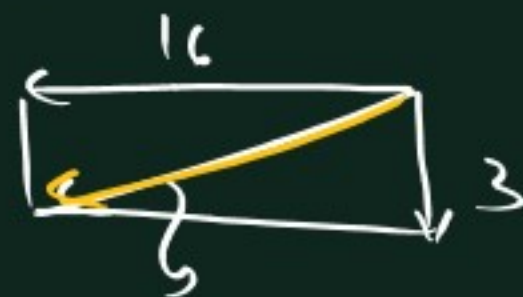
$$\begin{aligned}(\vec{A} + \vec{B}) &= (-16 + 0)\hat{i} \\ &\quad + (12 - 15)\hat{j} \\ &= -16\hat{i} - 3\hat{j}\end{aligned}$$

$$\begin{aligned}\cos 37^\circ &= 4/5 \\ \sin 37^\circ &= 3/5\end{aligned}$$

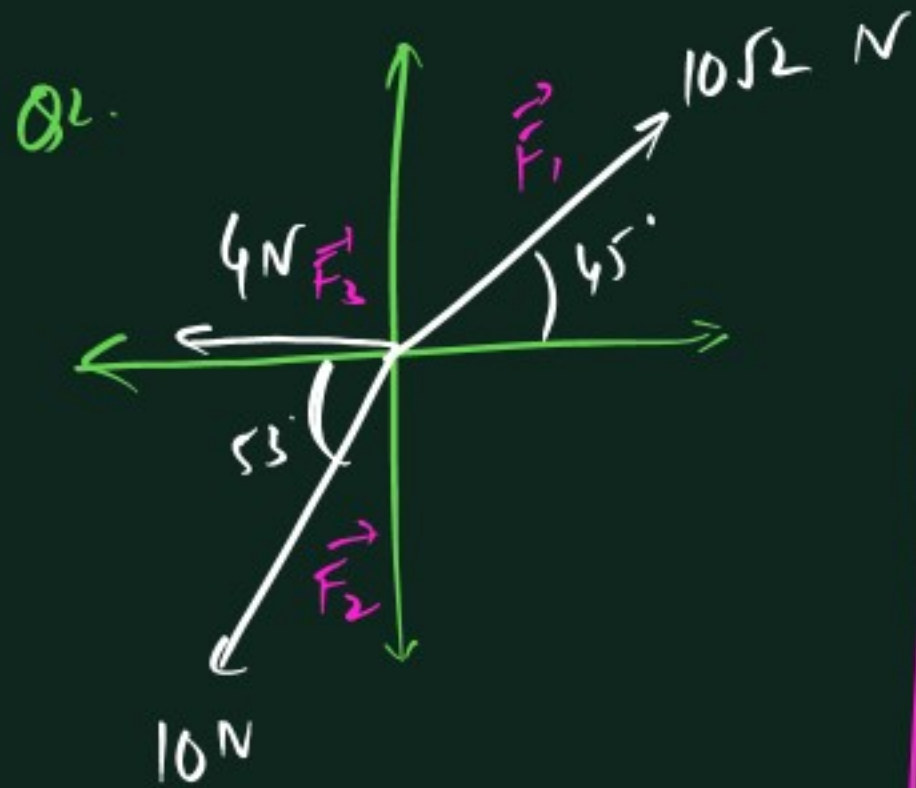
$$\cos 53^\circ = 3/5$$

$$\sin 53^\circ = 4/5$$

(iv)  $|\vec{A} + \vec{B}| = ?$

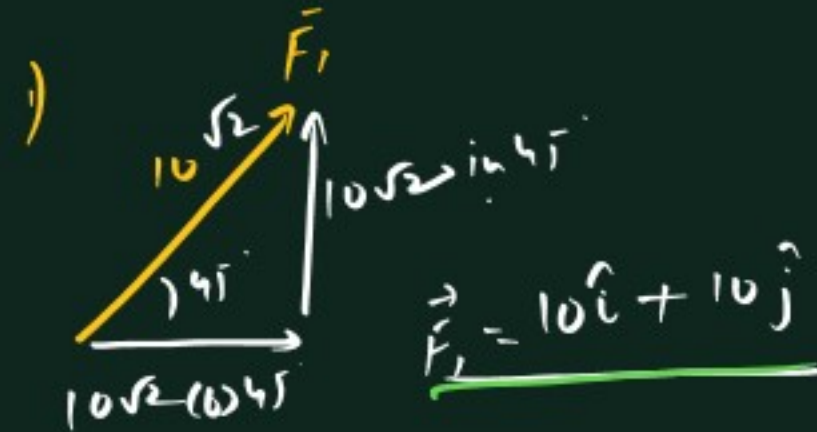


$$\begin{aligned}&\sqrt{16^2 + 3^2} \\ &= \sqrt{265}\end{aligned}$$

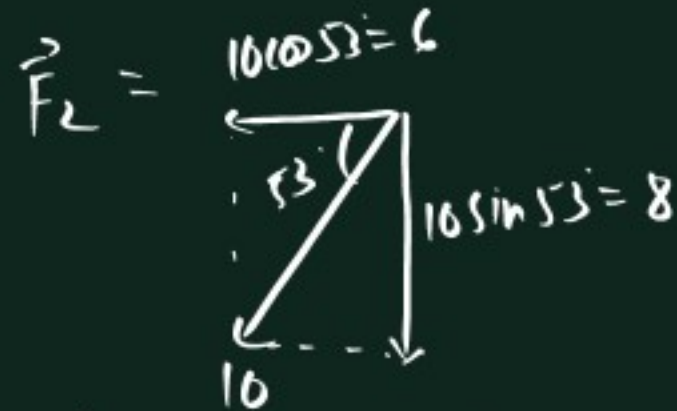


1) Find  $\vec{F}_1$ ,  $\vec{F}_2$ ,  $\vec{F}_3$  in  $(\hat{i}, \hat{j})$ .

2) Find net  $\vec{F}(\hat{i}, \hat{j})$  & its magnitude.



$$\vec{F}_1 = 10\hat{i} + 10\hat{j}$$



$$\vec{F}_2 = -6\hat{i} - 8\hat{j}$$

$$\vec{F}_3 = -4\hat{i}$$

$$\vec{F}_{\text{net}} = (10 - 6 - 4)\hat{i} + (10 - 8)\hat{j} = 2\hat{j}$$

$$|\vec{F}| = \sqrt{0^2 + 2^2} = 2 \text{ N}$$

Q3.  $\vec{A} = 8\hat{i} - 6\hat{j}$

Find unit vector in direction of  $\vec{A}$  ( $\hat{A}$ )

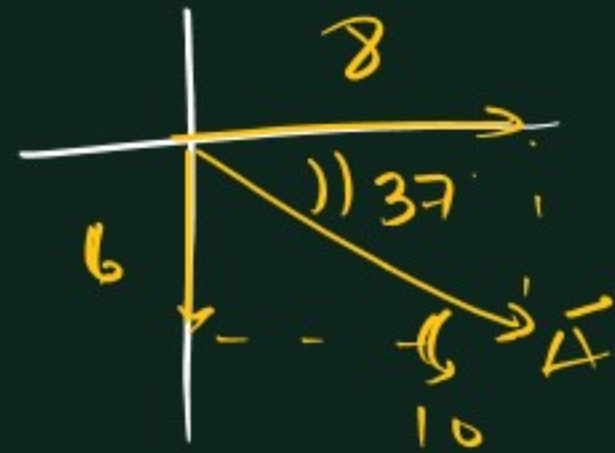
$$(\hat{i}, \hat{j})$$

$$\hat{A} = \frac{\vec{A}}{|\vec{A}|}$$

$$= \frac{8\hat{i} - 6\hat{j}}{\sqrt{8^2 + 6^2}}$$

$$= \frac{8}{10}\hat{i} - \frac{6}{10}\hat{j}$$

$$= 0.8\hat{i} - 0.6\hat{j}$$





Q1) If  $\hat{n} = \underline{k}\hat{i} + \underline{0.5}\hat{j}$   
 $\hat{n}$  = unit vector,  
 find value of  $k$ .

$$\underline{A} \quad |\hat{n}| = 1$$

$$= \sqrt{k^2 + (0.5)^2} = 1$$

$$\Rightarrow k^2 + \frac{1}{4} = 1$$

$$\Rightarrow k^2 = \frac{3}{4}$$

$$\Rightarrow \boxed{k = \pm \sqrt{3}/2}$$

Q5)  $\vec{A} = \underline{3}\hat{i} + \underline{2}\hat{j}$   
 $\vec{B} = 4\hat{i} - \hat{j}$

Find 1)  $2\vec{A} - \vec{B}$

2)  $|2\vec{A} - \vec{B}|$

$$1) \quad \begin{aligned} 2\vec{A} &= 6\hat{i} + 4\hat{j} \\ \vec{B} &= 4\hat{i} - \hat{j} \end{aligned}$$

$$2\vec{A} - \vec{B} = \underline{2\hat{i} + 5\hat{j}}$$

$$2) \quad |2\vec{A} - \vec{B}| = \sqrt{2^2 + 5^2} \\ = \underline{\underline{\sqrt{29}}}$$

Q6)  $\vec{A} = \underline{16}\hat{i} - \underline{12}\hat{j}$   
 $\vec{B} = 3\hat{i} + 4\hat{j}$

Find  $\vec{C}$  whose magnitude is same as  $\vec{A}$  and direction is along  $\vec{B}$ .

$$\underline{A} \quad \vec{C} = |\vec{C}| \hat{C}$$

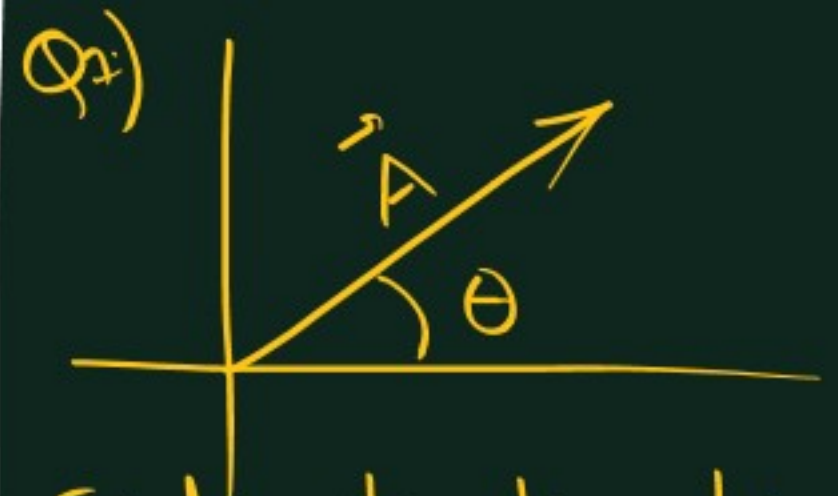
$\swarrow \quad \searrow$   
 $|\vec{A}| \quad \hat{B}$

$$\Rightarrow \vec{C} = |\vec{A}| \hat{B}$$

$$|\vec{A}| = \sqrt{16^2 + 12^2} = 20$$

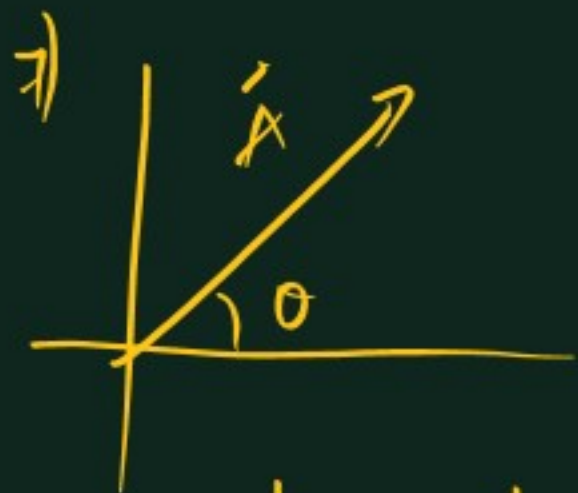
$$\hat{B} = \frac{\vec{B}}{|\vec{B}|} = \left( \frac{3\hat{i} + 4\hat{j}}{5} \right)$$

$$\vec{C} = \cancel{20} \left( \frac{3\hat{i} + 4\hat{j}}{\cancel{5}} \right) \\ = \underline{\underline{12\hat{i} + 16\hat{j}}}$$



Find unit vector along  $\vec{A}$ .



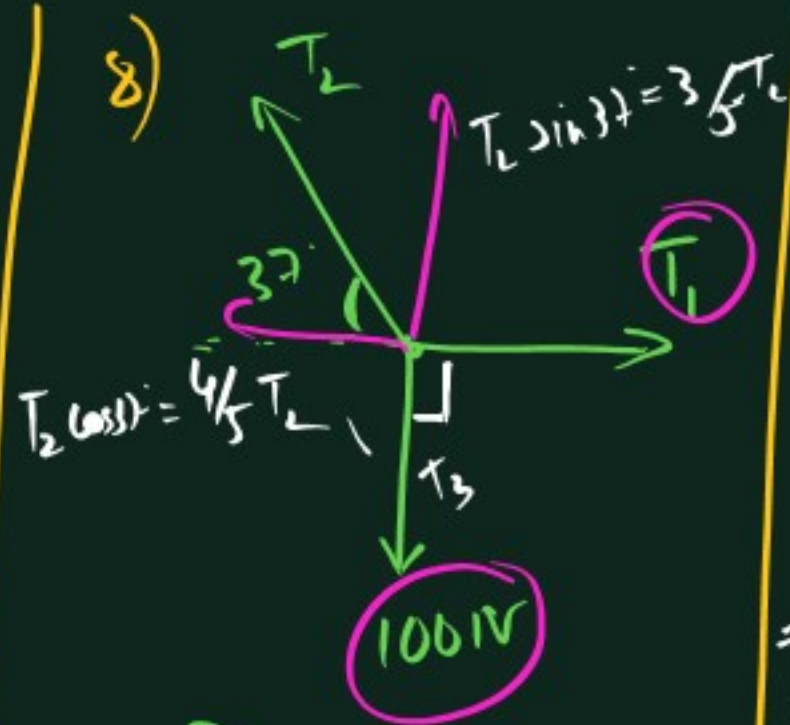


unit vector along  $\vec{A}$ ?

$$\hat{A} = \frac{\vec{A}}{|\vec{A}|}$$

$$= \frac{A \cos \theta \hat{i} + A \sin \theta \hat{j}}{A}$$

$$\hat{A} = \cos \theta \hat{i} + \sin \theta \hat{j}$$



Find  $T_1$  and  $T_2$   
if the 3 forces in  
equilibrium

$$\Sigma \vec{F} = 0$$

$$\vec{T}_1 = T_1 \hat{i}$$

$$\vec{T}_2 = -\frac{4}{5} T_2 \hat{i} + \frac{3}{5} T_2 \hat{j}$$

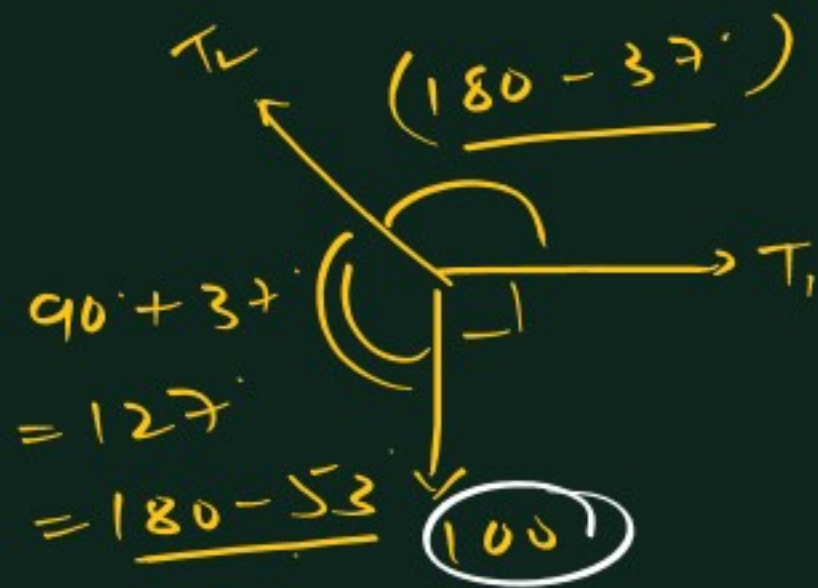
$$\vec{T}_3 = -100 \hat{j}$$

$$\vec{T}_1 + \vec{T}_2 + \vec{T}_3 = 0$$

$$= (T_1 - \frac{4}{5} T_2) \hat{i} + (\frac{3}{5} T_2 - 100) \hat{j} = 0$$

$$\begin{cases} T_1 - \frac{4}{5} T_2 = 0 \\ \frac{3}{5} T_2 - 100 = 0 \end{cases}$$

$$T_2 = \frac{500}{3}, T_1 = \frac{400}{3} \text{ N}$$



$$\frac{T_1}{\sin(180 - 53)} = \frac{T_2}{\sin 90}$$

$$= \frac{100}{\sin(180 - 37)}$$

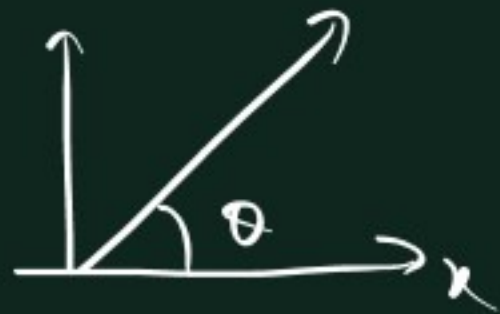
$$\Rightarrow \frac{T_1}{\sin 53} = \frac{T_2}{1} = \frac{100}{\sin 37}$$

$$\Rightarrow T_1 = \frac{400}{3}, T_2 = \frac{500}{3}$$



$$\sin(180 - \theta) = \sin \theta$$

$$\cos(180 - \theta) = -\cos \theta$$



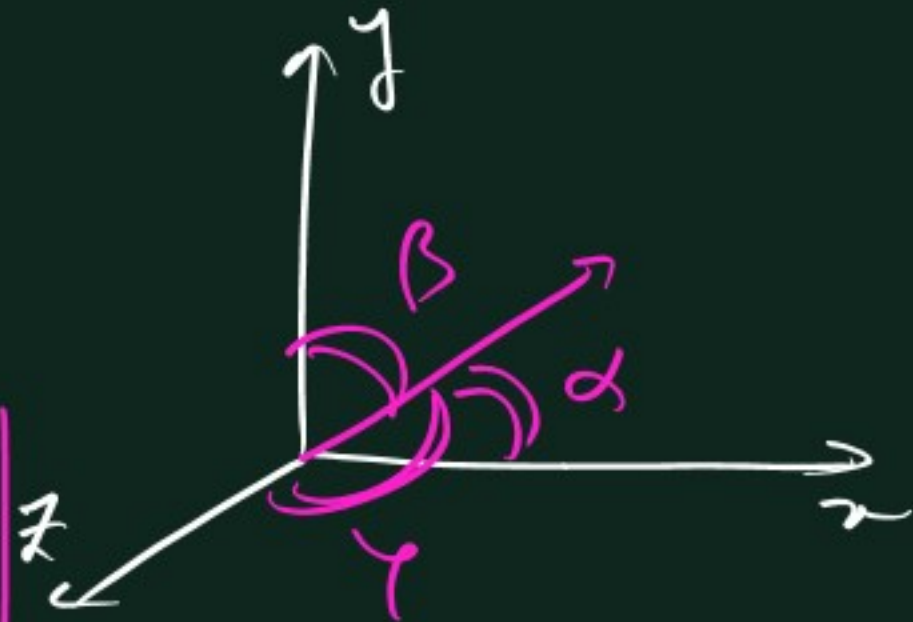
$$\cos \theta \hat{i} + \sin \theta \hat{j}$$

$$\begin{cases} \sin(90 + \theta) = \cos \theta \\ \cos(90 - \theta) = \sin \theta \\ \cos(90 + \theta) = -\sin \theta \end{cases}$$

$$\vec{A} = \underline{A_x} \hat{i} + \underline{A_y} \hat{j} + \underline{A_z} \hat{k}$$

$$\rightarrow |\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

$$\begin{cases} x\text{-axis, } \cos \alpha = \frac{A_x}{|\vec{A}|} \\ y\text{-axis, } \cos \beta = \frac{A_y}{|\vec{A}|} \\ z\text{-axis, } \cos \gamma = \frac{A_z}{|\vec{A}|} \end{cases}$$



$$\frac{\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma}{= 1}$$

9)  $\vec{A} = 4\hat{i} + 5\hat{j} - 3\hat{k}$

Find 1)  $|\vec{A}|$

2) Direction cosines of  $\vec{A}$

3)  $\hat{A}$  (unit vector)

1)  $|\vec{A}| = \sqrt{4^2 + 5^2 + 3^2}$   
 $= \sqrt{50} = 5\sqrt{2}$

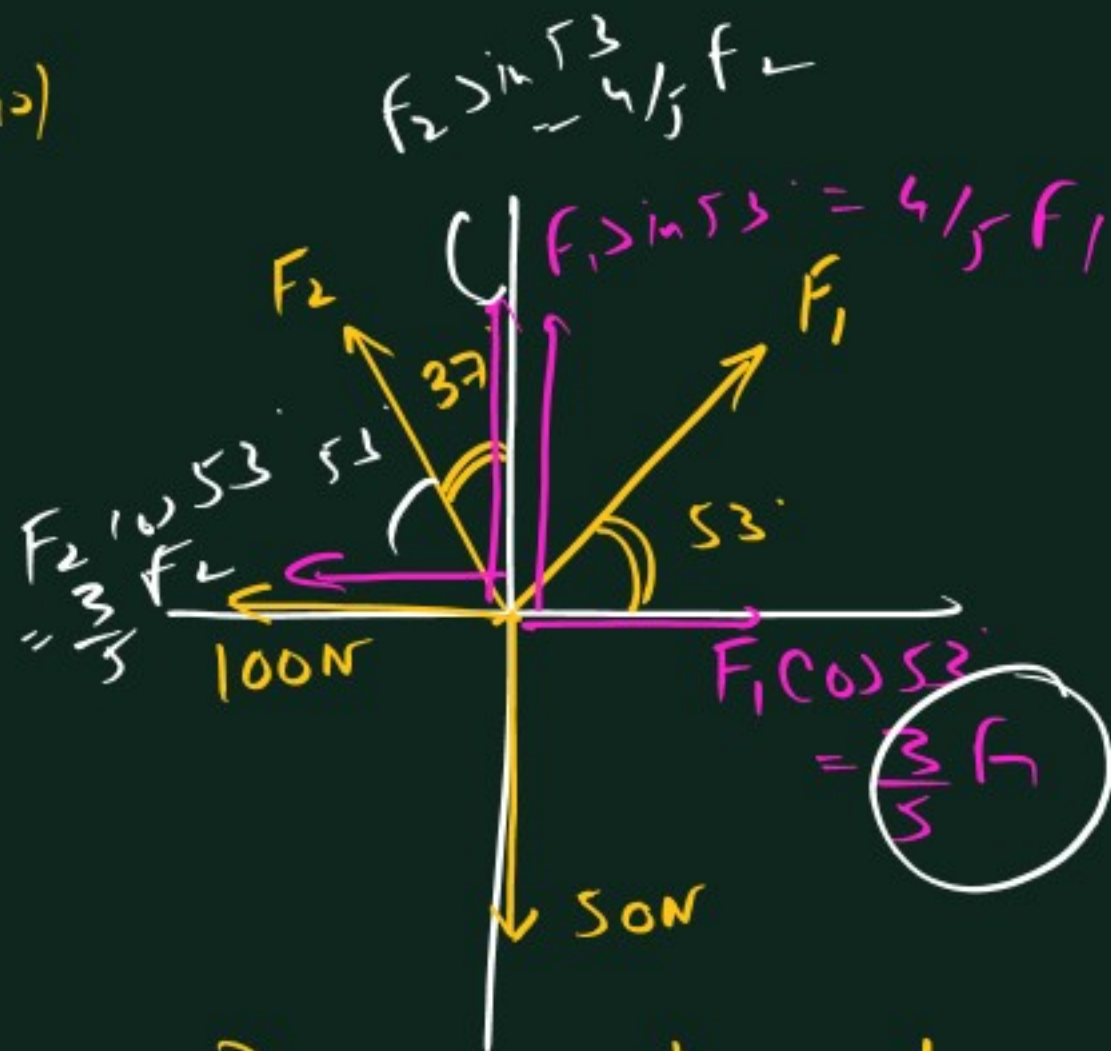
2)  $\cos \alpha = \frac{A_x}{A} = \frac{4}{5\sqrt{2}}$

$\cos \beta = \frac{A_y}{A} = \frac{5}{5\sqrt{2}} = \frac{1}{\sqrt{2}}$

$\cos \gamma = \frac{-3}{5\sqrt{2}}$

3)  $\hat{A} = \frac{\vec{A}}{|\vec{A}|} = \frac{4\hat{i} + 5\hat{j} - 3\hat{k}}{5\sqrt{2}}$

10)



Find  $F_1$  &  $F_2$  if all forces are in equilibrium

$$\vec{F}_{\text{net}} = \left( \frac{3}{5} F_1 - 100 - \frac{3}{5} F_2 \right) \hat{i}$$

$$+ \left( \frac{4}{5} F_2 + \frac{4}{5} F_1 - 50 \right) \hat{j}$$

$$\frac{3}{5} F_1 - 100 - \frac{3}{5} F_2 = 0$$

$$\Rightarrow 3F_1 - 3F_2 = 500$$

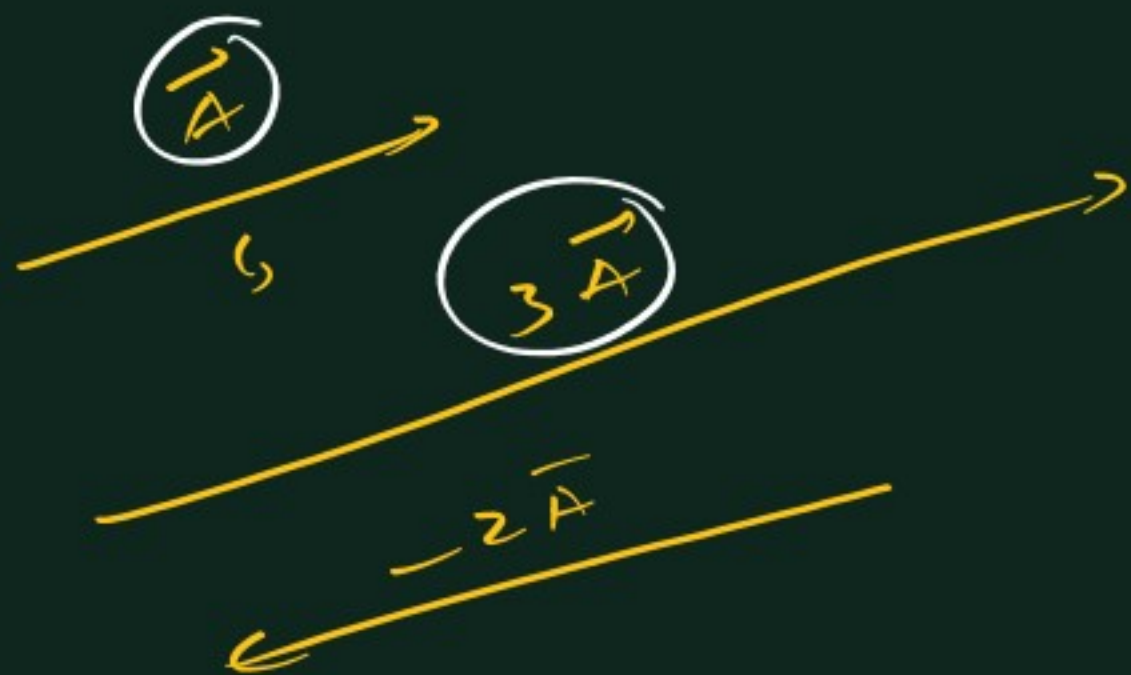
$$4F_1 + 4F_2 = 2500$$

$$2F_1 = \frac{500}{3} + \frac{250}{4}$$

$$F_1 = \frac{2750}{24} = 114.5$$

$$F_2 = -19N$$





$$\textcircled{Q} \quad \vec{A} = 4\hat{i} - \hat{j} + 2\hat{k}$$

$$\vec{B} = a\hat{i} + b\hat{j} + 6\hat{k}$$

If  $\vec{A} \parallel \vec{B}$ , Find  $a, b$ .

$$\vec{A} = 2\hat{i} + 3\hat{j} - \hat{k}$$

$$\vec{B} \text{ (circled)} = 6\hat{i} + a\hat{j} - 3\hat{k}$$

$$\frac{A_x}{B_x} = \frac{A_y}{B_y} = \frac{A_z}{B_z} = \textcircled{+} \text{ no}$$

$$\Rightarrow \frac{4}{a} = -\frac{1}{6} = \textcircled{\frac{2}{6}} \sim \frac{1}{3}$$

$$\Rightarrow a = 12, b = -3$$

$$\left( \begin{array}{l} \frac{4}{a} = \frac{1}{3} \\ -\frac{1}{6} = \frac{1}{3} \end{array} \right)$$



# Dot Product

## 1) Vector Multiplication

→ Dot Product  
→ Scalar product

→ Cross Product  
→ Vector Product

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$\theta = \text{angle b/w } \vec{A} \text{ \& } \vec{B}$

⑦  
⑤

$\vec{A} \cdot \vec{B} = 5 \times 7 \times \cos 60^\circ$   
 $= 35 \times \frac{1}{2} = 17.5$

3)  $\theta = \text{acute} \Rightarrow D.P. = \oplus$   
 $\theta = \text{obtuse} \Rightarrow D.P. = \ominus$

\*  $\theta = 90^\circ \Rightarrow D.P. = 0$

4)  $D.P. = \text{max}$  if  $\theta = 0$   
 $\vec{A} \cdot \vec{B} = AB$

$D.P. = \text{min}$  if  $\theta = 180$   
 $\vec{A} \cdot \vec{B} = -AB$

5)  $\vec{A} \cdot \vec{A} = AA \cos 0$   
 $= A^2$

$$\begin{cases} \vec{A} \cdot \vec{A} = A^2 \\ \therefore A = \sqrt{\vec{A} \cdot \vec{A}} \end{cases}$$

6)  $\hat{i} \cdot \hat{i} = 1$

$\hat{i} \cdot \hat{j} = 0$

$\hat{i} \cdot \hat{k} = 0$

$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$

7)  $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$

$\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$

1)  $\vec{A} = 4\hat{i}$

$\vec{B} = (2\hat{i} - \hat{j})$

Find  $(\vec{A} \cdot \vec{B})$

$4\hat{i} \cdot (2\hat{i} - \hat{j})$

$= 4\hat{i} \cdot 2\hat{i} - 4\hat{i} \cdot \hat{j}$

$= 8(\hat{i} \cdot \hat{i})$

$= 8$



$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

$$\text{Q. } \vec{A} = 2\hat{i} + \hat{j} - 4\hat{k}$$

$$\vec{B} = 3\hat{i} + 4\hat{j} + \hat{k}$$

$$\vec{A} \cdot \vec{B} = ??$$

$$\begin{aligned} \underline{\text{Ans}} \quad \vec{A} \cdot \vec{B} &= 2 \times 3 + 1 \times 4 + (-4) \times 1 \\ &= 6 \end{aligned}$$

Q. Find angle b/w  $\vec{A}$  &  $\vec{B}$

$$\underline{\text{Ans}} \quad \vec{A} \cdot \vec{B} = AB \cos \theta$$

$$\Rightarrow \cos \theta = \frac{\vec{A} \cdot \vec{B}}{AB}$$

$$= \frac{6}{\sqrt{2^2+1^2+4^2} \sqrt{3^2+4^2+1^2}}$$

$$\Rightarrow \cos \theta = \frac{6}{\sqrt{21} \sqrt{21}} = \frac{6}{\sqrt{546}}$$

$$\Rightarrow \theta = \cos^{-1} \left( \frac{6}{\sqrt{546}} \right)$$

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{AB}$$

$$\vec{A} \cdot \vec{B} = 0$$

$$\text{Q. } \vec{A} = 3\hat{i} + 7\hat{j} - 2\hat{k}$$

$$\vec{B} = 4\hat{i} + x\hat{k}$$

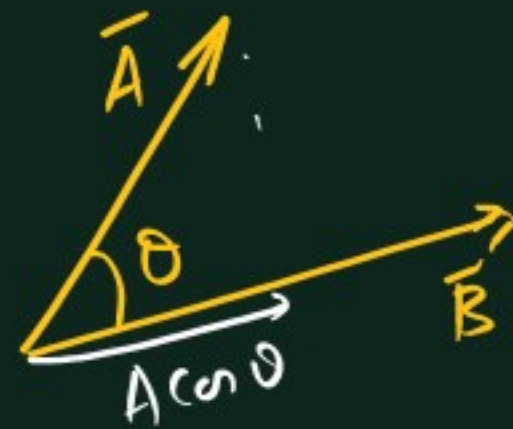
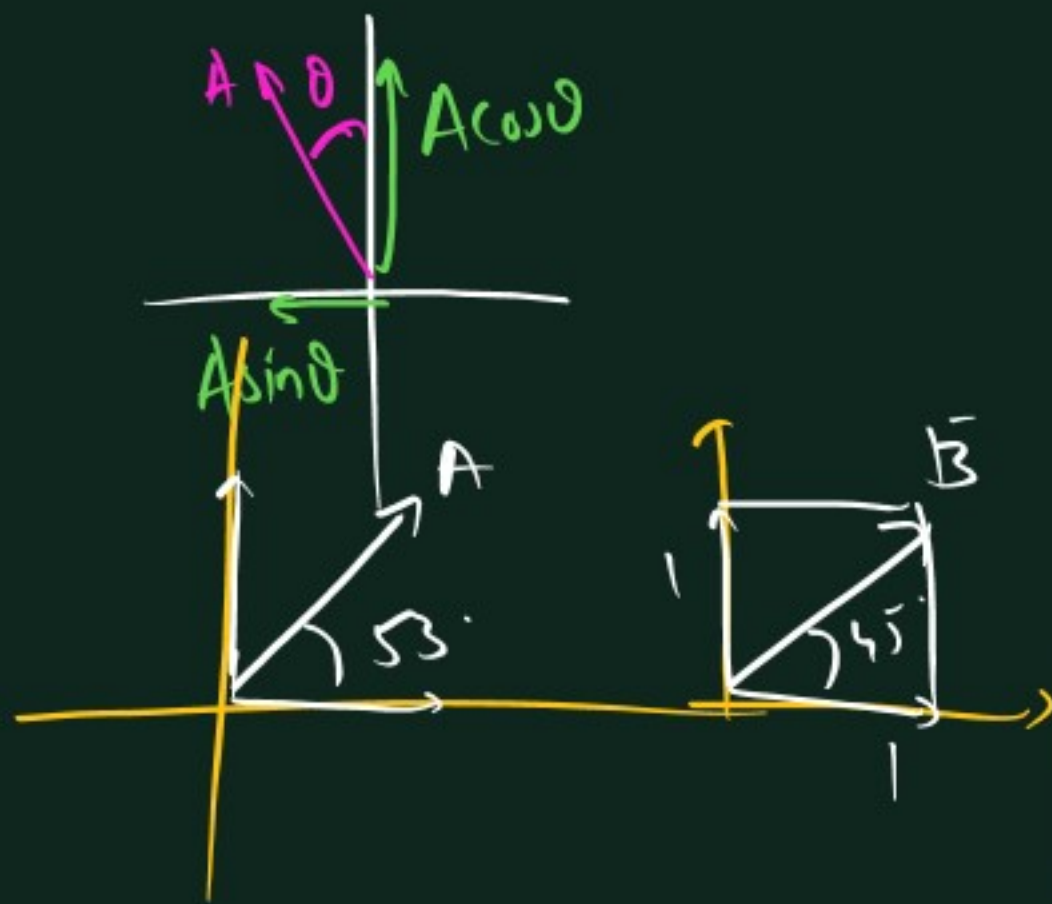
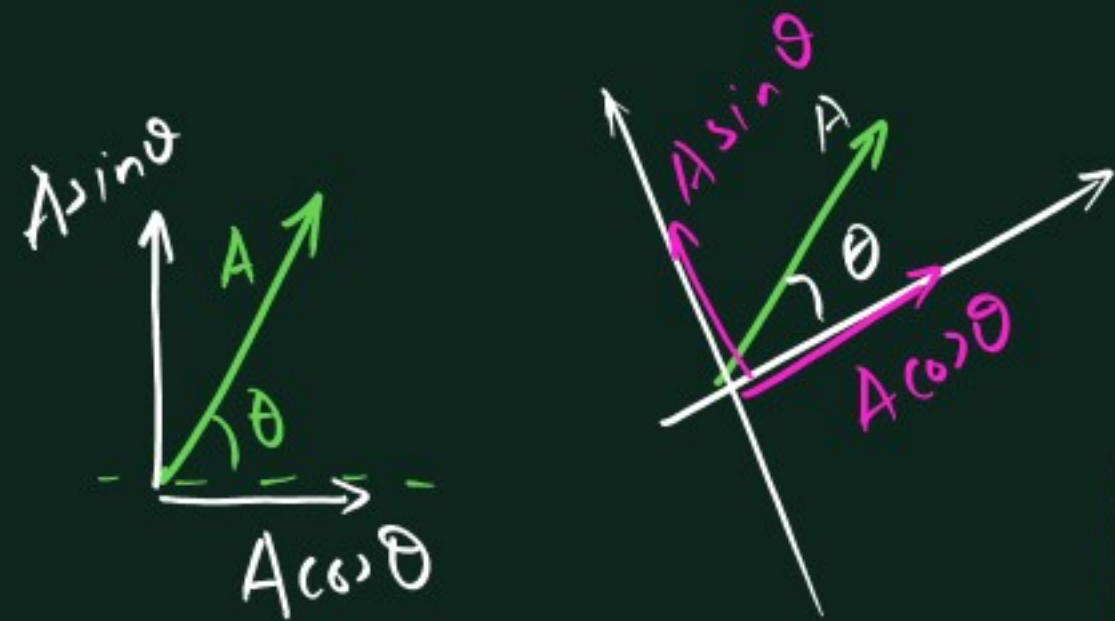
Find  $x$  if  $\vec{A} \perp \vec{B}$

$$\underline{\text{Ans}} \quad \vec{A} \cdot \vec{B} = 0$$

$$\Rightarrow 3 \times 4 + 0 + (-2)x = 0$$

$$\Rightarrow 12 = 2x \Rightarrow \boxed{x = 6}$$





To find component of  $\vec{A}$  along  $\vec{B}$

$$\Downarrow$$

$$A \cos \theta$$

$$= A \left( \frac{\vec{A} \cdot \vec{B}}{AB} \right)$$

$$= \frac{\vec{A} \cdot \vec{B}}{B}$$

$$\approx \vec{A} \cdot \hat{B}$$

$$Q \vec{A} = 3\hat{i} + 4\hat{j}$$

$$\vec{B} = \hat{i} + \hat{j}$$

Projection of  $\vec{B}$  along

$$\vec{A} = ??$$

$$A_{\text{proj}} = \vec{B} \cdot \hat{A}$$

$$\text{or}$$

$$\frac{\vec{B} \cdot \vec{A}}{|\vec{A}|}$$

$$= \frac{3 + 4}{5} = \left( \frac{7}{5} \right)$$



$$\vec{F} = 4\hat{i} - 2\hat{j} + 2\hat{k}$$

$$\vec{s} = \hat{i} + \hat{j} - \hat{k}$$

---

$$\begin{aligned}\vec{F} \cdot \vec{s} &= 4 \times 1 + (-2) \times 1 + 2 \times (-1) \\ &= 0\end{aligned}$$

$$(\text{Work} = \vec{F} \cdot \vec{s})$$

2. Three forces  $\vec{F}_1 = (3\hat{i} + 2\hat{j} - \hat{k})\text{N}$ ,  $\vec{F}_2 = (3\hat{i} + 4\hat{j} - 5\hat{k})\text{N}$  and  $\vec{F}_3 = A(\hat{i} + \hat{j} - \hat{k})\text{N}$  act simultaneously on a particle. In order that the particle remains in equilibrium, the value of  $A$  should be
- (A) ~~-6~~ (B) 6  
(C) 9 (D) -9

4. The magnitude of resultant of two equal forces is equal to either of two forces. The angle between forces is
- (A)  $\frac{\pi}{3}$  (B)  ~~$\frac{2\pi}{3}$~~   
(C)  $\frac{\pi}{2}$  (D)  $\frac{3\pi}{4}$

6. The two vectors  $\vec{A}$  and  $\vec{B}$  that are parallel to each other are
- (A)  ~~$\vec{A} = 3\hat{i} + 6\hat{j} + 9\hat{k}$ ,  $\vec{B} = \hat{i} + 2\hat{j} + 3\hat{k}$~~   
(B)  $\vec{A} = 3\hat{i} - 6\hat{j} + 9\hat{k}$ ,  $\vec{B} = \hat{i} + 2\hat{j} + 3\hat{k}$   
(C)  $\vec{A} = \hat{i} + 2\hat{j} + 3\hat{k}$ ,  $\vec{B} = \hat{i} + 2\hat{j} - 3\hat{k}$   
(D)  $\vec{A} = 3\hat{i} + 6\hat{j} - 9\hat{k}$ ,  $\vec{B} = \hat{i} - 2\hat{j} - 3\hat{k}$

15. 12 coplanar forces (all of equal magnitude) maintain a body in equilibrium, then the angle between any two adjacent forces is
- (A)  $15^\circ$  (B)  ~~$30^\circ$~~   
(C)  $45^\circ$  (D)  $60^\circ$

34. The vector that is added to  $(\hat{i} - 5\hat{j} + 2\hat{k})$  and  $(3\hat{i} + 6\hat{j} - 7\hat{k})$  to give a unit vector along the  $x$ -axis is

- (A)  $3\hat{i} + \hat{j} + 5\hat{k}$  (B)  $\hat{i} + 3\hat{j} + 5\hat{k}$   
(C)  ~~$-3\hat{i} - \hat{j} + 5\hat{k}$~~  (D)  ~~$3\hat{i} + \hat{j} - 5\hat{k}$~~

38. If  $\vec{a} = 2\hat{i} - 3\hat{j} + \hat{k}$  and  $\vec{b} = 3\hat{i} + \hat{j} - 2\hat{k}$ , the cosine of angle  $\theta$  between them is equal to

- (A)  $\frac{5}{14}$  (B)  $\frac{1}{7}$   
(C)  $\frac{3}{14}$  (D)  ~~$\frac{1}{14}$~~

$$\vec{A} + (\hat{i} - 5\hat{j} + 2\hat{k}) + (3\hat{i} + 6\hat{j} - 7\hat{k}) =$$



42. How many minimum number of coplanar vectors having different magnitudes can be added to give zero resultant.

- (A) 2 (B) 3  
(C) 4 (D) 5



43. How many minimum number of vectors in different planes can be added to give zero resultant?

- (A) 2 (B) 3  
(C) 4 (D) 5

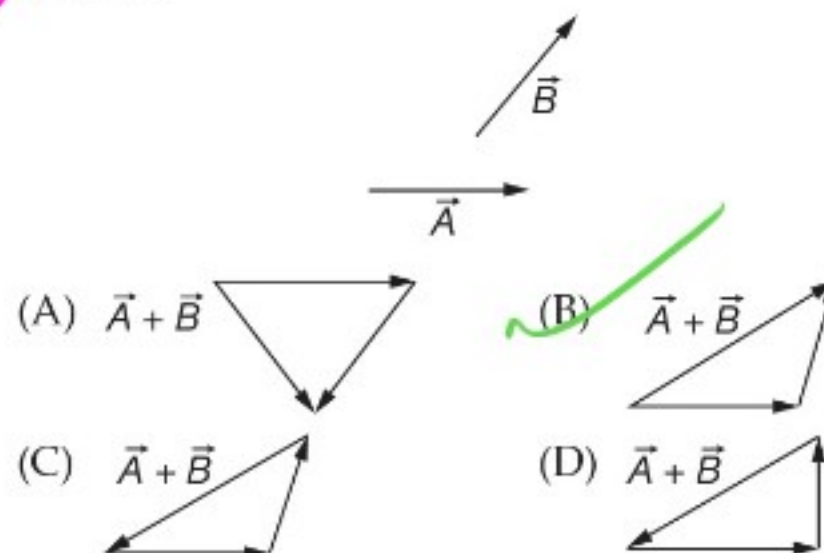
52. The maximum and minimum magnitudes of the resultant of two given vectors are 17 unit and 7 unit respectively. If these two vectors are at right angles to each other, the magnitude of their resultant is

- (A) 14 (B) 16  
(C) 18 (D) 13

60. The sum of the magnitudes of two vectors is 18 and the magnitude of their resultant is 12. If the resultant is perpendicular to one of the vectors, then what are the magnitudes of the two vectors?

- (A) 5, 13 (B) 6, 12  
(C) 7, 11 (D) 8, 10

73. Vectors  $\vec{A}$  and  $\vec{B}$  are shown in figure then diagram of  $\vec{A} + \vec{B}$  is

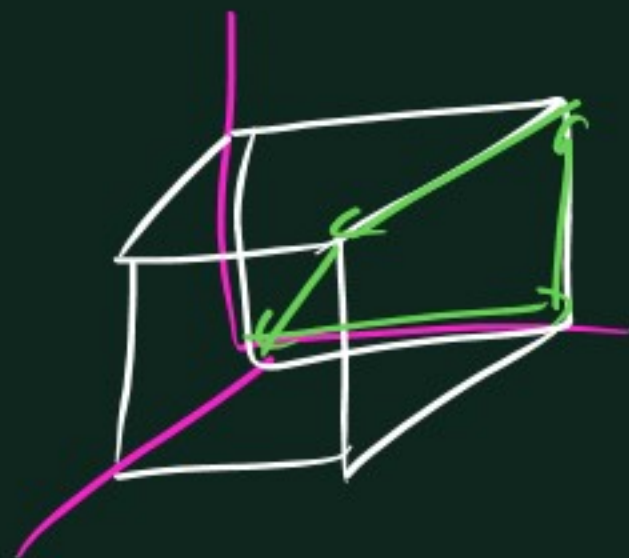


74. The resultant of two forces, one double the other in magnitude, is perpendicular to the smaller of the two forces. The angle between the two forces is

- (A) 120° (B) 60°  
(C) 90° (D) 150°

90. If  $\vec{A} = \hat{i} + 2\hat{j} + 2\hat{k}$  and  $\vec{B} = 3\hat{i} + 6\hat{j} + 2\hat{k}$ , then the vector in the direction of  $\vec{A}$  and having same magnitude as  $|\vec{B}|$ , is

- (A)  $\frac{7}{3}(\hat{i} + 2\hat{j} + 2\hat{k})$  (B)  $7(\hat{i} + 2\hat{j} + 2\hat{k})$   
(C)  $\frac{3}{7}(\hat{i} + 2\hat{j} + 2\hat{k})$  (D)  $\frac{7}{9}(\hat{i} + 2\hat{j} + 2\hat{k})$



$$\vec{C} = |\vec{B}| \hat{A}$$

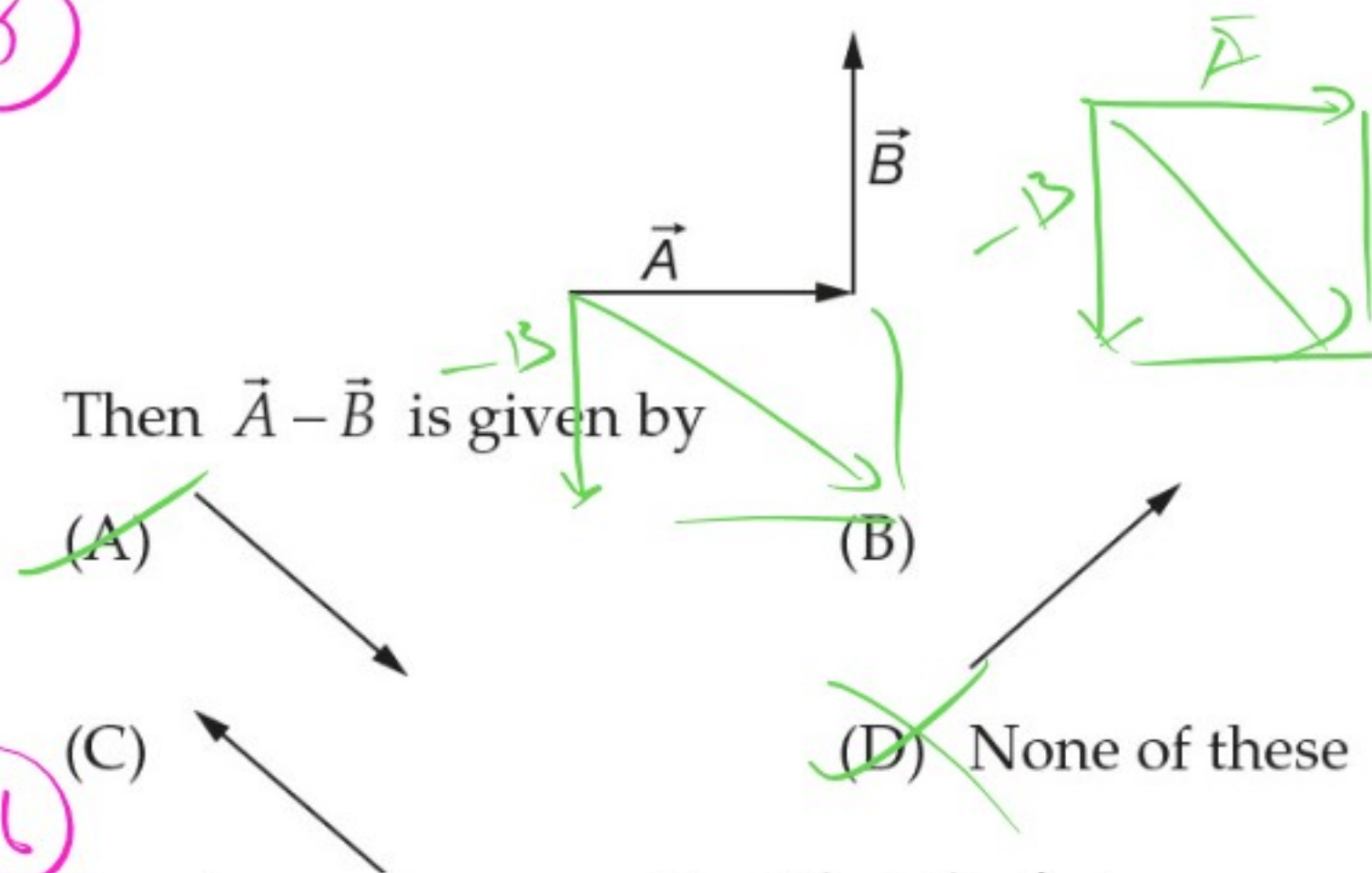
$$= \sqrt{3^2 + 6^2 + 2^2} \frac{(\hat{i} + 2\hat{j} + 2\hat{k})}{\sqrt{3^2 + 6^2 + 2^2}}$$

$$= \frac{7}{3}(\hat{i} + 2\hat{j} + 2\hat{k})$$



88. Which of the following sets of displacements might be capable of returning a car to its starting point?
- (A) 4, 6, 8 and 15 km (B) 10, 30, 50 and 120 km  
(C) 5, 10, 30 and 50 km (D) 50, 50, 75 and 200 km

97. Two vectors  $\vec{A}$  and  $\vec{B}$  are shown in the figure



116. If a unit vector is represented by  $0.5\hat{i} + 0.8\hat{j} + c\hat{k}$ , then the value of  $c$  is
- (A) 1 (B)  $\sqrt{0.11}$   
(C)  $\sqrt{0.01}$  (D)  $\sqrt{0.39}$

100. If a vector  $(2\hat{i} + 3\hat{j} + 8\hat{k})$  is perpendicular to the vector  $-4\hat{i} + 4\hat{j} + \alpha\hat{k}$ , then the value of  $\alpha$  is
- (A) -1 (B)  $\frac{1}{2}$   
(C)  $-\frac{1}{2}$  (D) 1

101. The unit vector parallel to the resultant of the vectors  $\vec{A} = 4\hat{i} + 3\hat{j} + 6\hat{k}$  and  $\vec{B} = -\hat{i} + 3\hat{j} - 8\hat{k}$  is

- (A)  $\frac{1}{7}(3\hat{i} + 6\hat{j} - 2\hat{k})$  (B)  $\frac{1}{7}(3\hat{i} + 6\hat{j} + 2\hat{k})$   
(C)  $\frac{1}{49}(3\hat{i} + 6\hat{j} + 2\hat{k})$  (D)  $\frac{1}{49}(3\hat{i} + 6\hat{j} - 2\hat{k})$

102. Following sets of three forces act on a body. Whose resultant cannot be zero?

- (A) 10, 10, 10 (B) 10, 10, 20  
(C) 10, 20, 20 (D) 10, 20, 40

117. The component of vector  $A = 2\hat{i} + 3\hat{j}$  along the vector  $\hat{i} + \hat{j}$  is

- (A)  $\frac{5}{\sqrt{2}}$  (B)  $10\sqrt{2}$   
(C)  $5\sqrt{2}$  (D) 5



